SAD Saturation — Original RTL

VCS + Verdi nWave | Directed 16×16 high-contrast corner case

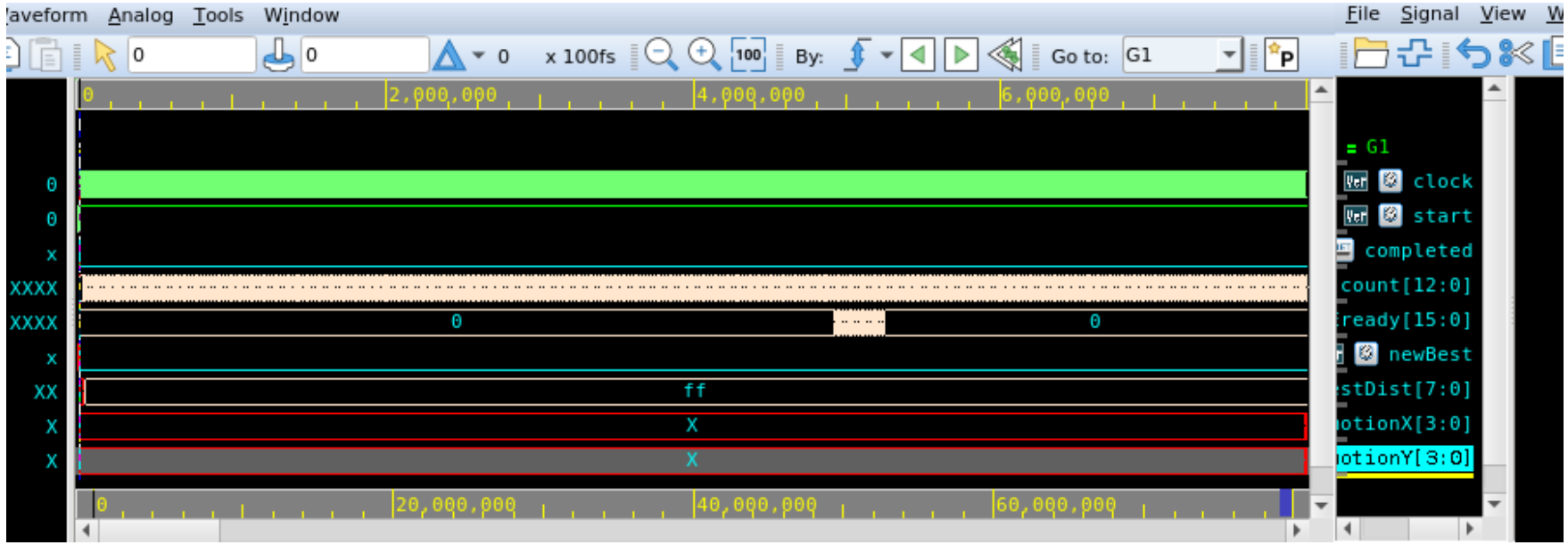


Figure 1: Original 8-bit datapath; full VCS result: completed = 1, BestDist = FF, motionX/Y = X.

Stimulus

Reference pixels = 00; search pixels = FF.

Expected 16-bit SAD

256 × |00 − FF| = 65,280 = FF00.

Observed failure

8-bit output saturates at FF; no newBest event.

Cause: the comparator resets BestDist to FF and accepts a candidate only when newDist < BestDist. FF cannot beat FF.

16-bit SAD — Verification After RTL Change

VCS + Verdi nWave | Separate full-precision and completion-hold variant

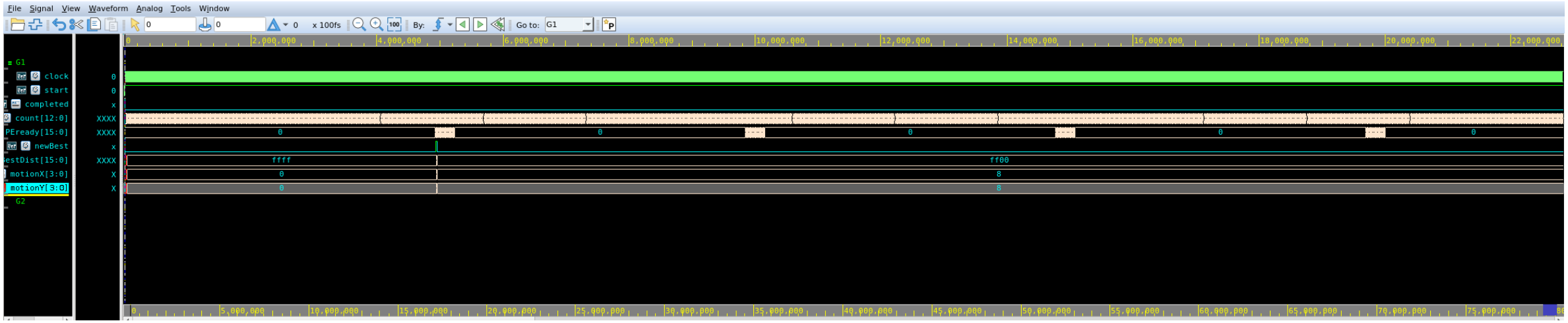


Figure 2: Synopsys FSDB shown in Verdi nWave; newBest asserts and BestDist becomes FF00.

BestDist

FF00 (65,280 decimal), 16-bit accumulator.

Selected vector

motionX=8, motionY=8 → (−8,−8).

Tie handling

All search candidates equal; first is retained.

Original professor RTL retained separately. The wider accumulator fixes the FF no-selection corner case; further varied-image regression is still required.

520 MHz ASIC Implementation

Synopsys Fusion Compiler X-2025.06-SP2a | SAED14 RVT library

Measured mapped-design comparison

Metric	Original 8-bit	16-bit V3B
Standard-cell area	881.96 µm²	1,193.03 µm²
Total mapped cells	1,693	1,905
Sequential cells	280	416
Macro / memory cells	0	0

V3B compile-stage setup slack: +0.06 ns (MET).

Area increased by 311.07 µm² (35.3%) versus the 8-bit baseline. This is the area cost of the corrected wider datapath; not post-route area.

Required course constraints

Clock period: 1.923 ns (520 MHz)

Input and output delay: 0.5 ns

Clock uncertainty: 0.2 ns

Reference/search memories: testbench only

Implementation checkpoint

Synthesis, floorplan and placement completed. CTS / route / DRC / extraction / PrimeTime final setup and hold are not yet verified.